CHRONOS-Photometry: Onboard Asteroid Lightcurve Extraction & Orbit Perturbation Tracker

ESA OSIP Call for Ideas: Autonomous Software Experiments on Hera | Category 6 – Open Innovation & Planetary Science

Proposal ID:	ESA-OSIP-HERA-2026-RDAS-ASTRO-CHRONOS	**Target Core:**	GR712RC LEON3 Core 1 (Bare-Metal @ 50 MHz)
Proposing Entity:	radixal s.r.o. (Brno, Czech Republic)	**Memory Limit:**	64 kB Stack | 28.6 kB Static RAM (0 malloc)
Leadership Triad:	Bc. Viktor Loš■ák (PI), Ing. Petr Slepi■ka, Mgr. David Riedl	**Standards:**	ECSS-E-ST-40C Cat D | MISRA-C:2012 Deterministic

EXECUTIVE SUMMARY & KEY IN-FLIGHT BUDGETS:
CHRONOS-Photometry is an onboard astronomical aperture photometry engine running on Core 1 bare-metal C. It extracts high-precision integrated lightcurves and mutual eclipse/occultation timings of Dimorphos and Didymos directly from Asteroid Framing Camera (AFC) images in real time.

- **CPU Utilization:** 3.6% @ 50 MHz SPARC V8 (Peak WCET: 0.85 s / 1020×1020 AFC frame)
- **RAM Footprint:** 28.6 kB Static RAM | < 10.0 kB Stack (Zero dynamic allocation / No malloc)
- **Timing Precision:** +/- 1.5 seconds for mutual eclipse/occultation event ingress/egress
- **Downlink Telemetry:** < 15.0 kB total data per 3-hour session (a 99.8% bandwidth reduction)
- **Scientific Legacy:** Direct synergy with the Astronomical Institute of the Czech Academy of Sciences (Ond■ejov).

1. Problem Statement & Deep-Space Constraints

Following NASA's DART kinetic impact, Dimorphos's orbital period was shortened by ~33 minutes, accompanied by predicted non-principal axis rotation (chaotic tumbling). Verifying this requires dense photometric lightcurve sampling. Ground telescopes are hindered by diurnal cycles and weather, while downloading full 1020×1020 raw images to reconstruct lightcurves on Earth requires gigabytes of downlink bandwidth—far exceeding Hera's 12 MB limit.

2. Technical Solution & Algorithmic Architecture

CHRONOS performs aperture photometry and harmonic curve inversion directly on board:

- **Stage 1 (Dynamic Synthetic Aperture Masking):** Calculates center-of-light for Didymos and Dimorphos, integrating flux within adaptive circular apertures.
- **Stage 2 (Photometric Normalization):** Calibrates instrumental flux against background stars and subtracts dark noise using integer math.
- **Stage 3 (Harmonic Eclipse Inversion):** Fits a fixed-point Fourier harmonic series to detect mutual eclipse ingress/egress timings.
- **Stage 4 (Ultra-Compact PUS Serialization):** Packages timestamped flux datapoints into 16-byte PUS Science Packets (APID 0x485).

4. Technical Feasibility & In-Flight Resource Budget

Resource Parameter	Hera Allocation / Limit	CHRONOS Consumption	Margin
CPU Clock & Execution	50 MHz SPARC V8 (Core 1)	0.85 s WCET per frame (3.6% load)	+96.4% CPU Idle
Stack Depth	64.0 kB max (at 0x40010000)	< 10.0 kB peak stack	+84.4% Stack Margin
Static RAM (BSS+Data)	Bounded Sandbox RAM	28.6 kB Static RAM (0 malloc)	Zero Heap Frag.
Downlink Data Volume	12.0 MB / session budget	< 15.0 kB total science telemetry	-99.8% Bandwidth Savings
Period Determination	Ground lightcurve accuracy	+/- 1.5 seconds in-situ precision	Sub-second Fidelity

5. Industrial Implementation Roadmap & Leadership Triad

Proposing Entity: Established in 2016 in Brno, Czech Republic, **radixal s.r.o.** specializes in mission-critical embedded systems, safety-critical railway controls (AK Signal / SIL standards), air-gapped defense architectures (URC Systems), national transport backbones (CENDIS / MD ■R), and commercial C-based optical satellite image processing in Norway.
• **Bc. Viktor Loš■ák (Principal Investigator & Lead Architect):** Over 10 years of embedded systems architecture and mathematical algorithm design. Responsible for overall pipeline design and ESA interface compliance.
• **Ing. Petr Slepi■ka (Engineering Lead & Delivery Director):** Specialist in safety-critical C software, MISRA-C static verification, automated QEMU CI/CD test harness, and ECSS Cat D qualification.
• **Mgr. David Riedl (Executive Director & Governance):** Responsible for contract governance, institutional compliance with ESA rules, and resource allocation.

6. References & Scientific Baseline

[1] **Pravec, P., Scheirich, P., et al. (Ond■ejov Observatory)**, *„Photometric survey of binary near-Earth asteroids and spin states of the Didymos system“*, Icarus, 2024.
[2] **Carnelli, I., et al.**, *„The ESA Hera Mission: Detailed Characterization“*, Advances in Space Research, 2022.
[3] **Harris, A. W., et al.**, *„Asteroid Lightcurve Parameters“*, Icarus.